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31 December 1971
SUPERSEDING
MIL-H-14844A(MU)
6 April 1970

MILITARY SPECIFICATION

HEAD ASSEMBLY: 8619900

1. SCOPE

1.1 This specification covers a modular unit of the Tank Commander's Periscopes, M34 or M36 herein referred to as Head Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

C8591108	Adapter, Torque
F11732111	Gage, Interchangeability, Maximum

FSC 1240

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F11732112	Gage, Interchangeability, Minimum
F8565556	Pressure Tester - Two Station
F8565964	Test Fixture for Periscopes M34 and M36
F8575658	Vibration Adapter for Periscopes M34 and M36
F8619900	Head Assembly: 8619900

PACKAGING DATA SHEET

8619900	Packaging of Head Assembly: 8619900

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- The materials shall be in accordance with the drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The head assembly shall be manufactured in accordance with Drawing F8619900, "Head Assembly" and drawings pertaining thereto.

3.2.1 Interchangeability.- The head assembly cavity shall be capable of the insertion and removal of the gages specified in 4.6.10.

3.3 General specification.- The contractor shall be responsible for adherence to, and compliance with the following requirements of MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.4 Environmental.

3.4.1 Storage temperature.- The head assembly shall show no evidence of cement separation, glass breakage or other physical failure, and shall meet all the requirements of this specification at standard ambient temperature (plus 60° to 90°F) after having been exposed and thermally stabilized at ambient temperatures of minus 80° to plus 160°F.

3.4.2 Operating temperatures.- The head assembly shall meet the applicable requirements of 3.7.1.2.2 while exposed and thermally stabilized at ambient temperatures of minus 40° and plus 150°F.

3.4.3 Vibration.- The head assembly shall show no evidence of physical failure due to being vibrated in a vertical plane at a constant frequency of 30 cycles per second (cps) with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds.

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3.5 Cleanliness.

a. Defects.- There shall be no evidence of moisture, grease, fingerprints, condensates, fractures, and adhesive separations on any glass component when viewing through either end.

b. Dirt.- There shall be no more than 3 particles of dirt or foreign material (classed as dirt) on any glass component. The size of any dirt particles shall not exceed 0.05 inch.

3.6 Sealing.- The head assembly shall show no evidence of leakage in excess of 0.2 pound per square inch (psi) when subjected to an internal pressure of 5 ± 0.10 psi for a minimum of 2 hours. The interior atmosphere of the accepted head assembly shall contain nitrogen(having a dew point at least as low as minus 25°F at atmospheric pressure) between 0.25 and 0.50 pound per square inch gage (psig).

3.7 Performance.

3.7.1 Orientation.- With the head assembly securely mounted with Datum Plane "A" horizontal (datum planes shown on Drawing F8619900), and with the elevation input arm positioned as indicated on Drawing F8619900, the head assembly shall meet the following requirements.

3.7.1.1 Collimation.

3.7.1.1.1 Pre-vibration.- A beam of light perpendicular to Datum Plane "B" (Drawing F8619900) shall be reflected 90 degrees plus or minus 0.3 mil to Datum Plane "A".

3.7.1.1.1.1 Post-vibration.- The original collimation reading established in 3.7.1.1.1 shall not deviate more than 0.1 mil subsequent to meeting the requirements of 3.4.3.

3.7.1.1.2 Resolution.- The resolution shall be 10 seconds of arc or less throughout the entire aperture when viewed at zero degrees elevation with a telescope of at least 15 power and at least 1.8 inch aperture. Resolution consists of detectable line structure and proper line count in all four meridians at one focus setting.

3.7.1.1.3 Image tilt.- The image of a line, originally parallel to the intersection of Datum Planes "A" and "B", shall remain parallel to this same line within one degree of arc.

3.7.1.2 Elevation.

3.7.1.2.1 Elevation input shaft travel.- Rotation of the elevation input shaft shall be limited by contact with the travel stops only. The shaft shall permit the line of sight to rotate a minimum of 60 degrees elevation and 20 degrees depression (see 6.3).

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3.7.1.2.2 Torque.- The running torque required to rotate the elevation input shaft through maximum elevation and depression shall not exceed 10 pound-inches at standard ambient temperature or 13 pound-inches at extreme temperatures of minus 40 degrees and plus 150 degrees Fahrenheit (see 6.3).

3.7.1.2.3 Plumb travel.- After establishing coincidence with a plumb line at zero degrees elevation, the line of sight shall not deviate from the plumb line by more than 0.25 mil when the line of sight is elevated 15 degrees and depressed 10 degrees by means of the elevation input arm. The line of sight shall not deviate from the plumb line by more than 1.0 mil from 15 to 60 degrees elevation and 10 to 15 degrees depression.

3.7.1.2.4 Elevation accuracy.- The established angle between the line of sight through the reticle boresight mark and the line through the centers of the two bores of the elevation input arm shall remain constant within plus or minus 0.2 mil, excluding backlash, at any setting of the arm between 10 degrees depression and 15 degrees elevation, and within plus or minus 1.5 mils at any setting above 15 degrees elevation up to and including 60 degrees, and from 10 degrees depression up to an including 15 degrees depression. The elevation or depression angle is defined as the angle between the line of sight and the horizontal Datum Plane "A" of the instrument.

3.7.1.2.5 Backlash.- Backlash between the optical line of sight and the elevation input arm shall not exceed 0.2 mil at any setting of the arm.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three head assemblies plus three each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts

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and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232, and Section 5 herein. In addition to 100% inspection as specified in 4.3.2.2, the tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.2 100 Percent inspection.- The following requirements and related tests shall be subjected to 100% inspection:

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
1. Interchangeability	3.2.1	4.6.10
2. Resolution	3.7.1.1.2	4.6.8

NOTE: The tests in 4.3.2.2 shall be conducted at a temperature between +60° and +90°F.

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TABLE I - CLASSIFICATION OF DEFECTS

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 0.65% Defective		
101.	Vibration	3.4.3	4.6.3
102.	Collimation, post-vibration	3.7.1.1.1.1	4.6.6.1
103.	Cleanliness	3.5	4.6.4
104.	Sealing	3.6	4.6.5
105.	Collimation, pre-vibration	3.7.1.1.1	4.6.2.1
106.	Elevation input shaft travel	3.7.1.2.1	4.6.6.2
107.	Plumb travel	3.7.1.2.3	4.6.6.3
108.	Elevation accuracy	3.7.1.2.4	4.6.6.4
109.	Backlash	3.7.1.2.5	4.6.6.5
110.	Image tilt	3.7.1.1.3	4.6.7
111.	Torque	3.7.1.2.2	4.6.9
MINOR:	NONE		

NOTE: The tests of Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One head assembly shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.2	Applicable drawing - visual
302.	General specification	3.3	MIL-F-13926 - visual

4.4.2 Environmental.- Three (3) head assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III, and shall then meet the requirements and tests in Table I.

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TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperatures (-80° and +160°F)	3.4.1	4.6.1
304.	Torque, elevation input shaft (-40° and +150°F)	3.4.2, 3.7.1.2.2	4.6.2, 4.6.9

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Sealing.- The pressure testing equipment to test sealing of the head assembly shall conform to the internal pressure test requirement of Sheet 1 of Drawing F8565556, Pressure Tester-Two Station.

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4.5.2.3.2 Performance - head assembly.- Special testing equipment to test performance of the head assembly shall conform to the requirements of Drawing F8565964, Test Fixture for Periscope, M34 and M36.

4.5.2.3.3 Image tilt.- The test for image tilt shall be performed with the head assembly positioned on special testing equipment that conforms to the following requirements.

a. The testing equipment shall be capable of securely mounting the head assembly with Datum Plane "A" shown on Drawing F8619900, in a horizontal plane within 10 seconds. The necessary mounting features (keyways and locating surfaces) shall be incorporated into the testing equipment.

b. A device that is capable of solidly arresting the elevation input arm in the position indicated on Drawing F8619900, shall be a part of the test equipment.

c. A target projector collimator having a clear aperture of at least 2.0 inches shall be a part of the test equipment. The target projector collimator shall be positioned at the entrance window of the head assembly so that the optical axis of the collimator is in alignment with the line of sight emitting from the central aperture (geometric center) of the head prism. The reticle of the target projector collimator shall have a crossline target test pattern with tolerance lines for projection into the optical system of the head assembly. The tolerance lines shall be used to determine the extent of image tilt. The vertical reticle line of the target test pattern shall be adjusted plumb. The optical axis of the target projector collimator shall be aligned parallel with respect to the optical axis of the viewing autocollimator and autocollimated to Datum Plane "B" shown on Drawing F8619900 within 5 seconds. The target image shall be adjusted for infinity.

d. A penta prism having a deviation of ±5 seconds (calibrated) shall be a part of the test equipment. The penta prism shall be positioned so that the optical axis of the penta prism is in alignment with the line of sight emitting from the central aperture (geometric center) of the head assembly.

e. A viewing autocollimator shall be a part of the test equipment. The viewing autocollimator shall be positioned at the penta prism so that the optical axis of the autocollimator is in alignment with the line of sight emitting from the central aperture (geometric center) of the exit window of the head assembly. The reticle of the viewing autocollimator shall have a crossline reticle pattern. The vertical reticle line of the viewing collimator shall be adjusted plumb. The optical axis of the viewing autocollimator shall be autocollimated through the penta prism to Datum Plane "A" shown on Drawing F8619900 within 5 seconds.

f. All set-up and calibration equipment with complete provisions necessary for certification and proof of accuracy of the test equipment shall be provided and shall become part of the testing equipment.

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4.6 Test methods and procedures.

NOTE: Caution shall be exercised during the following tests to prevent Arm Assembly 8635116 and Arm 8619845 from striking hard against their respective travel stop when the elevation input shaft is rotated through its full range of travel.

4.6.1 Storage temperatures.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.1. The head assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see NOTE below) to -80°F. The head assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually raised (see NOTE below) to +160°F. The head assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature. (+60° to +90°F), and the head assembly removed from the test chamber. The head assembly shall then be subjected to the visual, tactile, and audible examination of 4.6.3. The head assembly shall conform to the requirements of 3.4.1.

NOTE: The rate of temperature change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling tests of 4.6.1 and 4.6.2.

4.6.2 Operating temperatures.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.2. The head assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see NOTE in 4.6.1) to -40°F. The head assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, and while at -40°F, the head assembly shall be subjected to the test in 4.6.9 to determine compliance with the applicable requirements of 3.7.1.2.2 for this temperature (-40°F). Upon completion of this test, the temperature of the test chamber shall be gradually raised (see NOTE in 4.6.1) to +150°F. The head assembly shall remain at this temperature for a period of 5 hours. At the completion of this 5 hour period and while at +150°F, the head assembly shall again be subjected to the test in 4.6.9 to determine compliance with the applicable requirements of 3.7.1.2.2 for this temperature (+150°F). Upon completion of this test, the temperature of the test chamber shall be gradually reduced (see NOTE in 4.6.1) to room ambient temperature (+60° to +90°F) and the head assembly removed from the test chamber. Head assemblies which have passed the above tests shall be subjected to the test in 4.6.2.1.

4.6.2.1 Collimation, pre-vibration.- Prior to the vibration test of 4.6.3, the head assembly shall be positioned as in 4.6.6 to insure conformance with 3.7.1.1.1. With the elevation input arm positioned as indicated on Drawing F8619900 and the sector lever of the fixture adjusted to the zero degree position indicated on the index plate, position a

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beamsplitter at the eyepiece of the viewing collimator in front of the test fixture. While viewing through the beamsplitter, observe the viewing collimator reticle with respect to its reflected image. The horizontal and vertical reticle lines of the real and reflected image shall not deviate beyond the tolerance specified in 3.7.1.1.1 as measured on the collimator reticle. Record this measurement for use in the subsequent test of 4.6.6.1. The head assembly shall then be subjected to the test in 4.6.3. Caution must be exercised during this transfer to prevent Arm Assembly 8635116 and Arm 8619845 from striking hard against their respective travel stop.

4.6.3 Vibration.- This test shall be performed on head assemblies procured as a spare part item. When the head assemblies are provided with a complete periscope, this vibration test will not be required when the complete periscope is to be vibrated. Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.4.3 shall be utilized for the vibration test. Unless otherwise specified, the test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The head assembly shall be clamped on the vibrating equipment by means of an adapter that conforms to Drawing F8575658, Vibration Adapter for Periscopes M34 and M36. The vibration motion shall be in the vertical plane for the time duration specified in 3.4.3. During this test, the input elevation shaft shall prevent Arm Assembly 8635116 and Arm 8619845 from contacting their respective travel stop. Immediately following the vibration cycle, and prior to the examination for physical defects resulting from poor workmanship and materials, the head assembly shall be subjected to the test in 4.6.6.1. Caution shall again be exercised during transfer of the head assembly as specified in 4.6.2.1. Subsequent to the test in 4.6.6.1, the head assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.4.3, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall be made for functioning of moving parts, for irregularities of movement, friction or looseness. An audible examination shall also be made of the head assembly for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a head assembly defective. Head assemblies which have passed the above examinations shall be subjected to the remaining tests specified in 4.6.4 through 4.6.9 inclusive.

4.6.4 Cleanliness.- The optical surfaces of the head assembly shall be inspected for cleanliness in accordance with MIL-O-13830. The head assembly shall conform to the cleanliness requirements of 3.5.

4.6.5 Sealing.- This portion of the test shall be performed utilizing a pressure gage capable of reading between 0.25 to 0.50 pound per square inch gage (psig) and a dew point tester. The following test sequence shall be adhered to. Check the internal pressure of the head assembly utilizing the pressure gage specified herein. Then, test the dew point of the interior atmosphere of the head assembly by means of the dew point tester. Interior atmosphere of the head assembly shall conform to the requirements of 3.6. Upon satisfactory completion of this portion of the test, internal pressurizing of the head assembly shall be performed with special testing equipment conforming to 4.5.2.3.1. Pressurize the head assembly with dry nitrogen gas, having a dew point at least as low as minus 25°F, to 5.0 ± 0.10 psig

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in accordance with the procedure for internal pressure testing on Drawing F8565556. Retain the applied pressure in the head assembly for a minimum of 2 hours. During the first hour, the input elevation shaft shall be rotated through one full excursion of its articulation range. At the end of the 2 hour period there shall be no evidence of leakage in excess of 0.2 psi as indicated on the pressure gage of the testing equipment, to determine compliance with 3.6. Upon satisfactory completion of the sealing test, the head assembly shall then be flushed with the dry nitrogen gas specified herein. The interior atmosphere of the head assembly shall be periodically sampled through the dew point tester until the atmosphere shows a dew point at least as low as minus 25°F. When this atmosphere is attained within the head assembly, the head assembly shall then be sealed at the pressure specified in 3.6.

4.6.6 Performance.- The following tests shall be performed with the head assembly positioned on special testing equipment specified in 4.5.2.3.2. Set-up of the test equipment and the head assembly under test shall be performed in accordance with the set-up and operating instructions which are included on the test equipment drawing.

4.6.6.1 Collimation, post-vibration.- Perform this test utilizing the procedure in 4.6.2.1. Record the collimation measurement. Any difference between the recorded measurement obtained in this test and the recorded measurement obtained in 4.6.2.1 shall not exceed the tolerance specified in 4.7.1.1.1.1.

4.6.6.2 Elevation input shaft travel.- Perform this test subsequent to the test in 4.6.6.1. Observe the collimator reticle and note the position of the boresight point (intersecting point of the horizontal and vertical reticle lines) of the real and reflected image. Disengage the sector lever from the zero degree elevation position on the index plate and raise the set-up gage of the test fixture to its uppermost position and lock in place to allow the elevation input shaft to rotate freely during this test and the tests of 4.6.6.3 to 4.6.6.5 inclusive. While observing the boresight point of the real and reflected image, rotate the elevation input shaft so that the line of sight is deflected from 60° above to 20° below the noted position of the boresight point established herein. Rotation of the elevation input shaft shall conform to the requirements of 3.7.1.2.1, as measured on the index plate of the test fixture. Upon satisfactory completion of this position of the test, rotate the coupling to its maximum elevation, then to its maximum depression stop position. While the coupling is at each stop, back off the coupling for at least 1/4 rotation and then again return the coupling to the stop. Perform this a minimum of 3 times. During these rotations, the coupling shall show no evidence of irregularity of movement (sponginess or friction) while approaching or reaching the stop. Contact with the stop shall be solid, such that a distinct metal click can be heard to verify that coupling rotation is being limited only by means of contact with the coupling stop pin and not by contact of other components (optical or mechanical) to be in compliance with 3.7.1.2.1.

4.6.6.3 Plumb travel.- This test can be performed at the same time that the test of 4.6.6.2 is performed. While observing the boresight point of the real and reflected image, rotate the elevation input shaft so that the line of sight is deflected from 60° above to 20° below the

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noted position of the boresight point established in 4.6.6.2. The boresight point of the real and reflected image shall not deviate beyond the applicable value specified in 3.7.1.2.3 for the positions indicated as measured on the collimator reticle (see 4.6.6.4 when performing this test).

4.6.6.4 Elevation accuracy.- Perform this test in conjunction with the test in 4.6.6.3 (plumb travel). The horizontal reticle lines of the real and reflected image shall not deviate beyond the applicable value specified in 3.7.1.2.4 for the positions indicated.

4.6.6.5 Backlash.- With backlash removed for direction of travel, rotate the elevation input shaft and place the sector lever of the fixture in the zero degree elevation position on the index plate. Observe the reticle of the collimator reticle and record the position of the boresight point of the reflected image, as measured on the collimator reticle. Disengage the sector lever from the zero degree elevation position on the index plate and rotate elevation input shaft approximately 15 degrees in the same direction of travel, then with backlash included, return the sector lever of the fixture to the zero degree elevation position on the index plate without overtravel. Record this position of the boresight point of the reflected image, as measured on the collimator reticle. Any difference between the recorded readings is the amount of backlash. Backlash shall not exceed the tolerance specified in 3.7.1.2.5.

NOTE: Should overtravel occur during rotation of the elevation input shaft, the test is nullified and shall be repeated.

4.6.7 Image tilt.- This test shall be performed with the head assembly positioned on special testing equipment specified in 4.5.2.3.3. Observe the reticle of the viewing autocollimator positioned in front of the penta prism, and establish one extremity of the viewing autocollimator vertical reticle line coincident to the vertical target line. The image of the vertical target line shall be parallel to the vertical reticle line of the viewing autocollimator within the tolerance specified in 3.7.1.1.3, as determined by the tolerance lines on the target image.

4.6.8 Resolution.- Perform this test utilizing an observation telescope conforming to the requirements of 3.7.1.1.2, a resolving power chart representing the angular subtense for the seconds of arc specified for resolution and a supporting medium capable of receiving the observation telescope as well as the head assembly to be tested. Construction and positioning of the resolving power chart shall be in compliance with the "Resolution Test" specified in MIL-O-13830 and shall contain four line sets as shown in Figure 1 therein. Place the head assembly on the supporting medium and position it on its side with Army Assembly 8635116 positioned in accordance with Drawing F8619900. Position the observation telescope at the exit aperture of the head assembly. The focus of the observation telescope shall be established at one best focus setting for viewing of the detectable line structure and proper line count in all four meridians

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of the resolving power chart. Resolution shall meet the requirements specified in 3.7.1.1.2 throughout the entire aperture.

4.6.9 Torque, elevation input shaft.- Standard torque measuring equipment capable of measuring the torque values specified in 3.7.1.2.2 shall be utilized for this test. An adapter conforming to Drawing C8591108 shall be affixed to the elevation input shaft. The running torque required to rotate the elevation input shaft through its full range of travel in each direction shall not exceed the value specified in 3.7.1.2.2 for the temperatures indicated.

4.6.10 Interchangeability.- This test shall be performed utilizing the maximum and minimum interchangeability gages in Drawings F11732111 and F11732112 respectively, to determine compliance with 3.2.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 8619900.

6. NOTES

6.1 Intended use.- The Head Assembly, 8619900 is a modular unit of the Tank Commander's Periscopes M34 or M36, primarily used in the fire control system of the 105mm Gun Tank M60 Series.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification
b. Applicable packaging data sheet
c. Selection of applicable levels of preservation, packaging and packing
d. Applicable stock number
e. Requirement for first article testing

6.3 Elevation shaft.- At no time shall the elevation shaft be allowed to strike hard against the travel stops.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A333

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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